College Algebra

MTH 131 | Section B and I | Fall 2012

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Algebra: Review & Preview

Order of operations

Order of operations

- Please: Parenthesis and brackets

Order of operations

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals

Order of operations

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals
- My: Multiplication

Order of operations

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals
- My: Multiplication
- Dear: Division

Order of operations

- **P**lease: Parenthesis and brackets
- **E**xcuse: Exponents and Radicals
- **M**y: Multiplication
- **D**ear: Division
- **A**unt: Addition

Order of operations

- **P**lease: Parenthesis and brackets
- **E**xcuse: Exponents and Radicals
- **M**y: Multiplication
- **D**ear: Division
- **A**unt: Addition
- **S**ally: Subtraction

Order of operations

$$2(4-6)^2 + \frac{8+1}{3} - 1$$

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals
- My: Multiplication
- Dear: Division
- Aunt: Addition
- Sally: Subtraction

Order of operations

- **P**lease: Parenthesis and brackets
- **E**xcuse: Exponents and Radicals
- **M**y: Multiplication
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$$
\begin{aligned}
&2(4-6)^2 + \tfrac{8+1}{3} - 1 \\
&= 2(4-6)^2 + \tfrac{(8+1)}{3} - 1
\end{aligned}
$$

Order of operations

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals
- My: Multiplication
- Dear: Division
- Aunt: Addition
- Sally: Subtraction

$$2(4-6)^2+\tfrac{8+1}{3}-1$$
$$=2(4-6)^2+\tfrac{(8+1)}{3}-1$$
$$=2(-2)^2+\tfrac{9}{3}-1$$

Order of operations

- **P**lease: Parenthesis and brackets
- **E**xcuse: Exponents and Radicals
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$$
\begin{aligned}
&2(4-6)^2 + \tfrac{8+1}{3} - 1 \\
&= 2(4-6)^2 + \tfrac{(8+1)}{3} - 1 \\
&= 2(-2)^2 + \tfrac{9}{3} - 1 \\
&= 2(4) + \tfrac{9}{3} - 1
\end{aligned}
$$

Order of operations

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\begin{aligned}
&2(4-6)^2+\tfrac{8+1}{3}-1\\
&=2(4-6)^2+\tfrac{(8+1)}{3}-1\\
&=2(-2)^2+\tfrac{9}{3}-1\\
&=2(4)+\tfrac{9}{3}-1\\
&=8+\tfrac{9}{3}-1
\end{aligned}
$$

Order of operations

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\begin{aligned}
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&=2(4-6)^2+\tfrac{(8+1)}{3}-1\\
&=2(-2)^2+\tfrac{9}{3}-1\\
&=2(4)+\tfrac{9}{3}-1\\
&=8+\tfrac{9}{3}-1\\
&=8+3-1
\end{aligned}
$$

Order of operations

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\begin{aligned}
&2(4-6)^2 + \tfrac{8+1}{3} - 1 \\
&= 2(4-6)^2 + \tfrac{(8+1)}{3} - 1 \\
&= 2(-2)^2 + \tfrac{9}{3} - 1 \\
&= 2(4) + \tfrac{9}{3} - 1 \\
&= 8 + \tfrac{9}{3} - 1 \\
&= 8 + 3 - 1 \\
&= 11 - 1
\end{aligned}
$$

Order of operations

- Please: Parenthesis and brackets
- Excuse: Exponents and Radicals
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- Sally: Subtraction

$$
\begin{aligned}
&2(4-6)^2+\tfrac{8+1}{3}-1\\
&=2(4-6)^2+\tfrac{(8+1)}{3}-1\\
&=2(-2)^2+\tfrac{9}{3}-1\\
&=2(4)+\tfrac{9}{3}-1\\
&=8+\tfrac{9}{3}-1\\
&=8+3-1\\
&=11-1\\
&=10
\end{aligned}
$$

Note that subtraction can be rewritten as addition.

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ex) $4 - 3 =$

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ex) $4 - 3 = 4 + (-3) = 1$

Also, division can be rewritten as multiplication.

ex) $\frac{9}{3} = \frac{1}{3} \cdot 9 = 3$

Note that subtraction can be rewritten as addition.

ex) $4 - 3 = 4 + (-3) = 1$

Also, division can be rewritten as multiplication.

ex) $\frac{9}{3} = \frac{1}{3} \cdot 9 = 3$

So the acronym could've been P.E.D.M.S.A.

FRACTIONS!!!!

ex) $\frac{1}{2} + \frac{3}{4} \neq \frac{1+3}{2+4}$

FRACTIONS!!!!

ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4}=\frac{4}{6}=\frac{2}{3}$

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ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

But why?

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You have to do the division/multiplication steps BEFORE the addition.

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You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1) + \frac{3}{4} = \frac{1}{2}(\frac{2}{2}) + \frac{3}{4}$

FRACTIONS!!!!

ex) $\frac{1}{2} + \frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

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You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1) + \frac{3}{4} = \frac{1}{2}\left(\frac{2}{2}\right) + \frac{3}{4} = \frac{2}{4} + \frac{3}{4}$

FRACTIONS!!!!

ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

But why?

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You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1) + \frac{3}{4} = \frac{1}{2}(\frac{2}{2}) + \frac{3}{4} = \frac{2}{4} + \frac{3}{4}$

Now, $\frac{2}{4} + \frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$

FRACTIONS!!!!

ex) $\frac{1}{2} + \frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

But why?

$\frac{1}{2} + \frac{3}{4} = \frac{1}{2}(1) + \frac{1}{4}(3)$

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Now, $\frac{2}{4} + \frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$

But why are we allowed to do this now?

FRACTIONS!!!!

ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

But why?

$\frac{1}{2}+\frac{3}{4} = \frac{1}{2}(1)+\frac{1}{4}(3)$

You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1)+\frac{3}{4} = \frac{1}{2}(\frac{2}{2})+\frac{3}{4} = \frac{2}{4}+\frac{3}{4}$

Now, $\frac{2}{4}+\frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$

But why are we allowed to do this now?

$\frac{2}{4}+\frac{3}{4}$

FRACTIONS!!!!

ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4}=\frac{4}{6}=\frac{2}{3}$

But why?

$\frac{1}{2}+\frac{3}{4}=\frac{1}{2}(1)+\frac{1}{4}(3)$

You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1)+\frac{3}{4}=\frac{1}{2}(\frac{2}{2})+\frac{3}{4}=\frac{2}{4}+\frac{3}{4}$

Now, $\frac{2}{4}+\frac{3}{4}=\frac{2+3}{4}=\frac{5}{4}$

But why are we allowed to do this now?

$\frac{2}{4}+\frac{3}{4}=\frac{1}{4}(2)+\frac{1}{4}(3)$

Now we <u>have</u> to do the addition first!

FRACTIONS!!!!

ex) $\frac{1}{2}+\frac{3}{4} \neq \frac{1+3}{2+4} = \frac{4}{6} = \frac{2}{3}$

But why?

$\frac{1}{2}+\frac{3}{4} = \frac{1}{2}(1)+\frac{1}{4}(3)$

You have to do the division/multiplication steps BEFORE the addition.

ex) $\frac{1}{2}(1)+\frac{3}{4} = \frac{1}{2}(\frac{2}{2})+\frac{3}{4} = \frac{2}{4}+\frac{3}{4}$

Now, $\frac{2}{4}+\frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$

But why are we allowed to do this now?

$\frac{2}{4}+\frac{3}{4} = \frac{1}{4}(2)+\frac{1}{4}(3) = \frac{1}{4}(2+3)$

Now we have to do the addition first!

Homework

HW pg. 86-87: 2, 24, 30, 32, 46, 90, 98, 110, 120